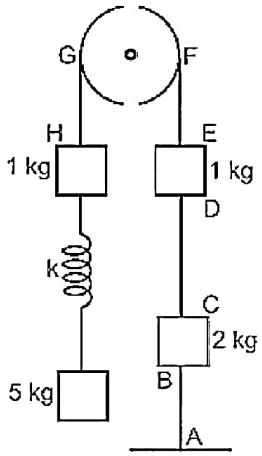
Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

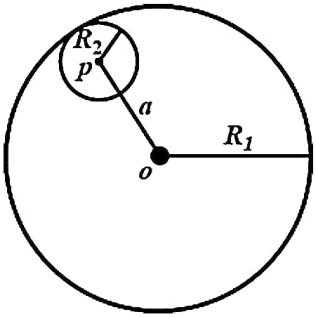
In the situation shown in the diagram, the system is in equilibrium. All the strings, spring and pulley are light. Just after cutting the string AB,



- The tension in the string GH is 6 g
- The tension in the string GH is zero
- The tension in string GH is $\frac{21g}{4}$
- The 2 kg block remains in equilibrium

Question No. 2 **(Only One Option Correct Type)**

Consider a uniform spherical charge distribution of radius R_1 centred at the origin O. In this distribution, a spherical cavity of radius R_2, centre at P with distance $OP = \alpha = R_1 - R_2$ is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is (are)



A. $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$

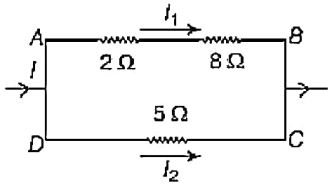
B. $\vec{E}$ is uniform, its magnitude is depends on R_2 and its direction depends on $\vec{r}$

C. $\vec{E}$ is uniform, its magnitude is independent of α but its direction depends on $\vec{\alpha}$

D. $\vec{E}$ is uniform and both its magnitude and direction depend on $\vec{\alpha}$

Question No. 3 **(Only One Option Correct Type)**

In the circuit shown, the heat produced in 5Ω resistance due to current through is 50 J/s. Then, the heat generated per second in 2Ω resistance is



- 5 J/s
- 4 J/s
- 9 J/s
- 10 J/s

Question No. 4 **(Only One Option Correct Type)**

Assertion (A): The upper surface of the wing of an aeroplane is made convex and the lower surface is made concave.
Reason (R): The air currents at the top have smaller velocity and thus less pressure at the bottom than at the top.

- Both A and R are true and R is a correct explanation for A
- Both A and R are true but R is not a correct explanation for A
- A is true, R is false
- A is false, R is false

Question No. 5 **(Only One Option Correct Type)**

$\frac{E^2}{\mu_0}$ has the dimensions ($E =$ electric field, $\mu_0 =$ permeability of free space)

- $\left[M^2L^3T^{-2}A^2\right]$

- $\left[\mathrm{MLT}^{-4}\right]$
- $\left[\mathrm{ML}^{3}\ \mathrm{T}^{-2}\right]$
- $\left[M^{-1}L^{2}TA^{-2}\right]$

Question No. 6 **(Only One Option Correct Type)**

An electron moves with speed $2 \times 10^{5}\ \mathrm{m/s}$ along the positive x-direction in the presence of a magnetic induction $B = \hat{i} + 4\hat{j} - 3\hat{k}$ (in tesla). The magnitude of the force experienced by the electron in newton is (charge on the electron $= 1.6 \times 10^{-19}\mathrm{C}$)

- 1.18×10^{-13}
- 1.28×10^{-13}
- 1.6×10^{-13}
- 1.72×10^{-13}

Question No. 7 **(Only One Option Correct Type)**

An installation consists of an electric motor which drives a water pump to lift $75\ \mathrm{L}$ of water per second to a height of $5\ \mathrm{m}$, where water is disbursed at negligible speed. If the motor consumes a power of $5\ \mathrm{kW}$, then what is the efficiency (%) of the installation? $[\mathrm{g} = 10\ \mathrm{m\ s^{-2}}]$

- 75
- 45
- 85
- 65

Question No. 8 **(Only One Option Correct Type)**

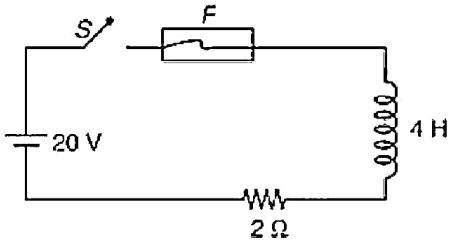
A closed organ pipe of length ' L ' and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. If the frequencies of their first overtones are same, then the length of the open organ pipe is

- $\frac{4L}{3}\sqrt{\frac{\rho_2}{\rho_1}}$
- $\frac{4L}{3}\sqrt{\frac{\rho_1}{\rho_2}}$
- $\frac{4L}{3}$

○ $\frac{L}{3}$

Question No. 9 **(Only One Option Correct Type)**

In the circuit shown below the battery and inductor are ideal. F is a fuse having zero resistance and will blow when the current through it reaches 5 A. The switch S is closed at $t = 0$, the fuse will blow at t equal to $\ln(x)$. The value of x is ?



○ 4

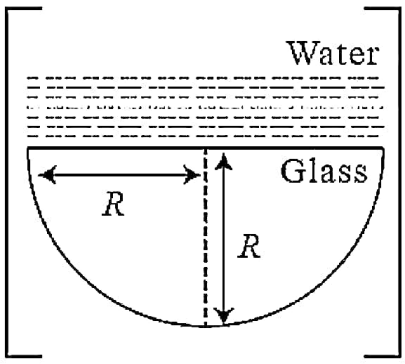
○ 3

○ 5

○ 2

Question No. 10 **(Only One Option Correct Type)**

A ray of light travelling in glass $\left(\mu = \frac{3}{2}\right)$ is incident on a horizontal glass air surface at the critical angle θ_c. If thin layer of water $\left(\mu = \frac{4}{3}\right)$ is now poured on the glass air surface, the angle at which the ray emerges into air the water-air surface is



○ 60°

○ 45°

○ 90°

D) 180°

Question No. 11 **(Only One Option Correct Type)**

The de Broglie wavelength of a charged particle accelerated through a potential difference V is λ. If the potential difference is increased by 21%, the de Broglie wavelength of the charged particle is

- $\frac{5\lambda}{9}$
- $\frac{7\lambda}{9}$
- $\frac{9\lambda}{11}$
- $\frac{10\lambda}{11}$

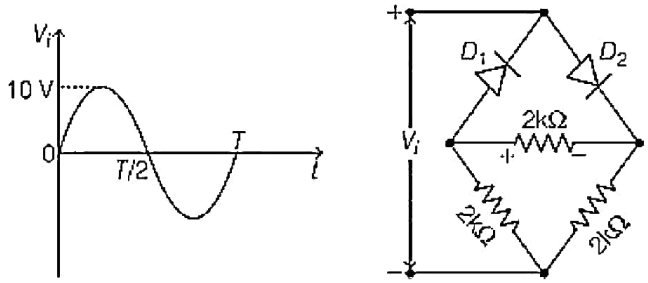
Question No. 12 **(Only One Option Correct Type)**

A conducing loop of resistance $10\ \Omega$ and area $3.5 \times 10^{-3}\ \text{m}^2$ is placed in uniform and time varying magnetic field $B = 0.4 \sin(50\pi t)$. The charge passing through the loop in $t = 0$ to $t = 10$ ms is: (mark answer in μC)

- 170
- 140
- 185
- 165

Question No. 13 **(Only One Option Correct Type)**

In the circuit shown for the given input voltage V_i, the maximum output voltage V_0 is



- 0
- 5 V
- 10 V
- $\frac{5}{\sqrt{2}}$ V

Question No. 14 **(Only One Option Correct Type)**

In the Young's double slit experiment, the intensities at two points P_1 and P_2 on the screen are respectively I_1 and I_2. If P_1 is located at the centre of a bright fringe and P_2 is located at a distance equal to a quarter of fringe width from P_1, then $\frac{I_1}{I_2}$ is____

- 2
- $\frac{1}{2}$
- 4
- 16

Question No. 15 **(Only One Option Correct Type)**

Using the following data

Mass of hydrogen atom $= 1.00783u$

Mass of neutron $= 1.00867\text{u}$

Mass of nitrogen atom $\left({}_7\text{N}^{14}\right) = 14.00307\text{u}$

The calculated value of the binding energy of the nucleus of the nitrogen atom $\left({}_7\text{N}^{14}\right)$ is close to

- 56MeV
- 98MeV
- 104MeV
- 112MeV

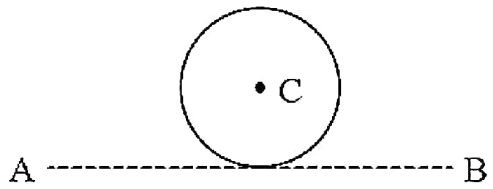
Question No. 16 **(Only One Option Correct Type)**

The angular velocity of the earth with which it has to rotate so that acceleration due to gravity on 60° latitude becomes zero is (Radius of the earth= 6400 km. At the poles $g = 10\ \text{m s}^{-2}$)

- $2.5 \times 10^{-3}\ \text{rad s}^{-1}$
- $5.0 \times 10^{-1}\ \text{rad s}^{-1}$
- $100\ \text{rad s}^{-1}$
- $7.8 \times 10^{-2}\ \text{rad s}^{-1}$

Question No. 17 **(Only One Option Correct Type)**

A thin wire of length l having density ρ is bent into a circular loop with C as its centre, as shown in figure. The moment of inertia of the loop about the line AB is



- $\frac{5\rho l^3}{16\pi^2}$
- $\frac{\rho l^3}{16\pi^2}$
- $\frac{\rho l^3}{8\pi^2}$
- $\frac{3\rho l^3}{8\pi^2}$

Question No. 18 **(Only One Option Correct Type)**

The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will

- Be independent of α and β
- Drop to zero when $\alpha = \beta$
- Go on decreasing with time
- Go on increasing with time

Question No. 19 **(Only One Option Correct Type)**

One mole of an ideal gas under goes a process $P = P_0\left[1 + \left(\frac{2V_0}{V}\right)^2\right]^{-1}$ where P_0, V_0 are constants. Change in temperature of the gas when volume is changed from $V = V_0$ to $V = 2V_0$ is:

- $\frac{4}{5}\frac{P_0V_0}{R}$
- $\frac{3}{4}\frac{P_0V_0}{R}$
- $\frac{2}{3}\frac{P_0V_0}{R}$
- $\frac{9}{7}\frac{P_0V_0}{R}$

Question No. 20 **(Only One Option Correct Type)**

Two slabs A and B of different materials but of the same thickness are joined end to end to form a composite slab. The thermal conductivities of A and B are k_1 and k_2 respectively. A steady temperature difference of $12^\circ\mathrm{C}$ is maintained across the composite slab. If $k_1 = \frac{k_2}{2}$, the temperature difference across slab A is

- $4^\circ\mathrm{C}$
- $6^\circ\mathrm{C}$
- $8^\circ\mathrm{C}$
- $10^\circ\mathrm{C}$

Physics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

A simple harmonic motion is represented by $\mathrm{y} = 0.1\sin(0.02\pi\mathrm{t})$. Find the minimum time taken by the particle to go directly from its extreme position (amplitude) to half of the amplitude. (t is in second) Mark answer to nearest integer in seconds.

Question No. 2 **(Numerical Type)**

5 g of ice at $-30^\circ\mathrm{C}$ and 20 g of water at $35^\circ\mathrm{C}$ are mixed together in a calorimeter.

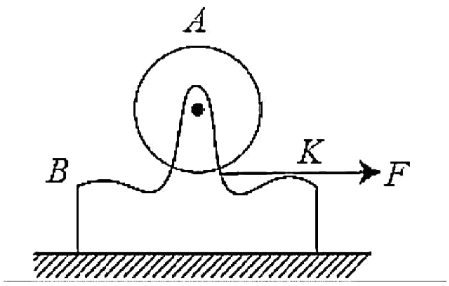
The final temperature of the mixture is (Neglect heat capacity of the calorimeter, specific heat capacity of ice $= 0.5\mathrm{calg}^{-1}{}^\circ\mathrm{C}^{-1}$ and latent heat of fusion of ice $= 80\mathrm{calg}^{-1}$ and specific hea: capacity of water $= 1\mathrm{calg}^{-1}{}^\circ\mathrm{C}^{-1}$)

Question No. 3 **(Numerical Type)**

A parallel plate capacitor of capacity C_0 is charged to a potential V_0. (i) The energy stored in the capacitor when the battery is disconnected and the plate separation is doubled is E_1. (ii) The energy stored in the capacitor when the charging battery is kept connected and the separation between the capacitor plates is doubled is E_2. Then, E_1/E_2 value is:

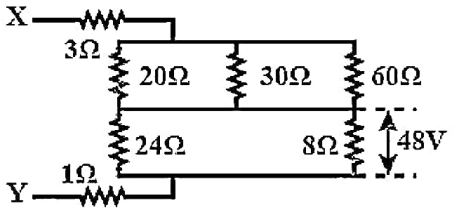
Question No. 4 **(Numerical Type)**

A uniform solid cylinder A of mass m_1 can freely rolate about a horizontal axis fixed to a mount of mass m_2, A constant horizontal force F is applied to the end K of a light thread tightly wound on the cylinder. The friction between the mount and the supporting horizontal plane is assumed to be absent. Take $m_1 = m_2 = 5$ kg and $\mathrm{F} = 10$ N, The acceleration of the point K is a m/s^2 then value of a



Question No. 5 **(Numerical Type)**

The potential difference across 8 Ω resistance is 48 V as shown in figure. The value of potential difference across points X and Y will be



Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

Which of the following is not a disproportionation reaction?.....

- ○ $Hg_2Cl_2 \longrightarrow Hg + HgCl_2$
- ○ $4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$
- ○ $Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$
- ○ $BaO_2 + H_2SP_4 \longrightarrow BaSO_4 + H_2O_2$

Question No. 2 **(Only One Option Correct Type)**

Assertion (A) : $AlCl_3$ exists as a dimer through halogen bridged bonds.
Reason (R) : $AlCl_3$ gets stability by accepting electrons from the bridged halogen

- ○ Both (A) and (R) are true and (R) is the correct explanation of (A)

- Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (A) is true, but (R) is not true
- (A) is not true, but (R) is true

Question No. 3 **(Only One Option Correct Type)**

In the detection of the nitrate ion, a brown complex in the form of a brown ring is formed at the junction of the two liquids. How many unpaired electrons does this brown complex contain?

- 2
- 1
- 0
- 3

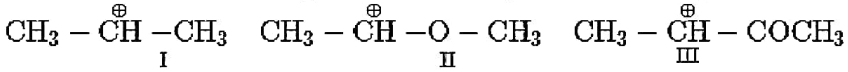
Question No. 4 **(Only One Option Correct Type)**

2 –Bromobutane (A) is treated with NaI in the presence of dry acetone to give compound 'B'. The compound 'B' is boiled with moist silver oxide to give compound 'C'. Identify the compounds 'B' and 'C'. How many optical isomers are possible for compound 'C'?

- 2 –Iodobutane, Butan–2 –ol, 2 isomers
- 3 –Iodopentane, Pentan–3 –ol, 4 isomers
- 2 –Iodobutane, Butan–2 –ol, 4 isomers
- 3 –Iodopentane, Pentan–3 –ol, 2 isomers

Question No. 5 **(Only One Option Correct Type)**

The CORRECT decreasing order of stability of the following carbocations is:



- I > II > III
- III > II > I
- II > III > I
- II > I > III

Question No. 6 **(Only One Option Correct Type)**

The hybridisation of metals involved in the following complexes, respectively are $\left[\mathrm{Mn(CN)_6}\right]^{3-}$, $[\mathrm{CoF_6}]^{3-}$

- sp^3d^2, sp^3d^2
- sp^3d^2, d^2sp^3
- d^2sp^3, d^2sp^3
- d^2sp^3, sp^3d^2

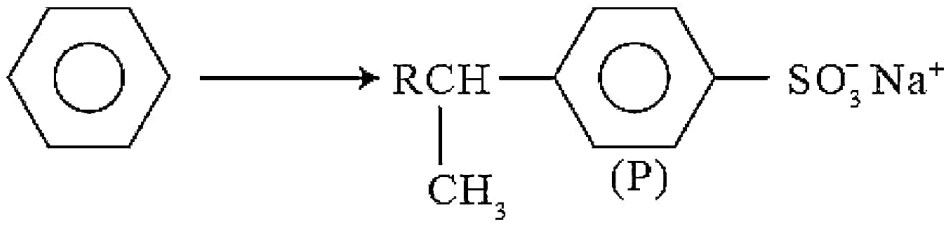
Question No. 7 **(Only One Option Correct Type)**

In Lassaigne's test for halogens, it is necessary to remove 'X' and 'Y' from the sodium fusion extract, if nitrogen and Sulphur are present. This is done by boiling the extract with 'Z'. Identify X, Y and Z.

- $\mathrm{X = NaNO_3\, Y = Na_2SO_4 Z =}$ conc. $\mathrm{HNO_3}$
- $\mathrm{X = NaNH_2\, Y = NaSH} Z =$ conc. HCl
- $\mathrm{X = NaNO_2\, Y = Na_2SO_3 Z =}$ conc. $\mathrm{H_2SO_4}$
- $\mathrm{X = NaCN}$
  $Y = \mathrm{Na_2S}$
  $\mathrm{Z =}$ conc. $\mathrm{HNO_3}$

Question No. 8 **(Only One Option Correct Type)**

For the preparation of a detergent 'P' from benzene, The following steps are involved:



I. $\xrightarrow[\text{reaction}]{\mathrm{RCH{=}CH_2, HF, Friedel\text{-}Crafts}}$

II. $\xrightarrow{\mathrm{H_2SO_4, SO_3}}$

III. $\xrightarrow{\mathrm{NaOH}}$

These steps should be in sequence of

- I, II, III
- II, I, III
- II, III, I

D) III, I, II

Question No. 9 **(Only One Option Correct Type)**

Number of incorrect statements are-

(A) The π bond between metal and carbonyl carbon reduces the bond order of C – O in carbon monoxide

(B) dz^2 orbital of central metal atom/ ion is used in dsp^2 hybridisation.

(C) CN^- is a π- acid Ligand.

(D) All negative ligands are stronger than neutral ligands.

○ 2

○ 1

○ 4

○ 3

Question No. 10 **(Only One Option Correct Type)**

Equivalent conductance of 1M CH_3COOH is 10 ohm^{-1} cm^2 $equiv^{-1}$ and that at infinite dilution is 200 ohm^{-1} cm^2, $equiv^{-1}$. Hence % ionisation of CH_3COOH is :

○ 5%

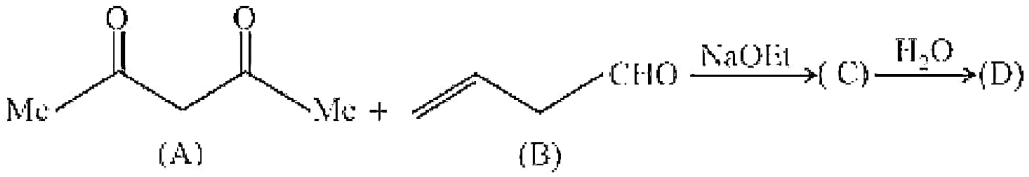
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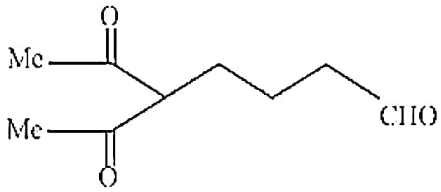
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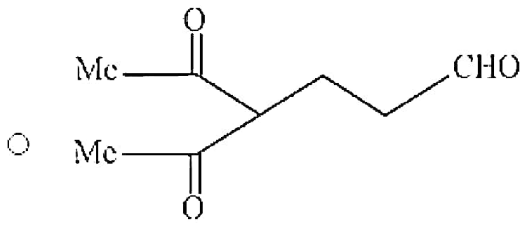
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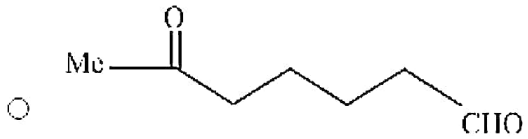
Question No. 11 **(Only One Option Correct Type)**

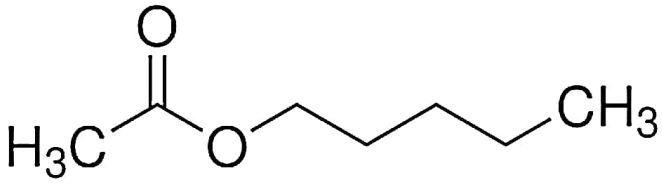
What is product D ?



○ 



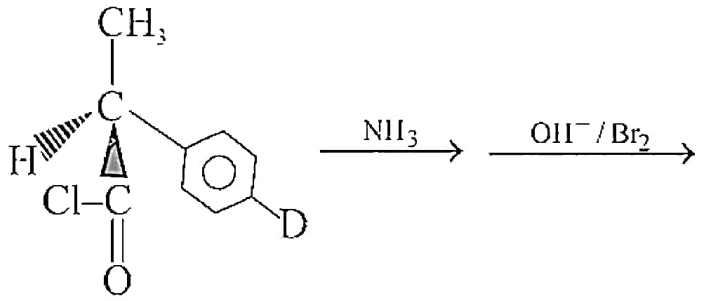


D) 

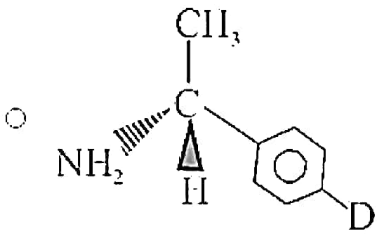
#add -CHO instead of CH3

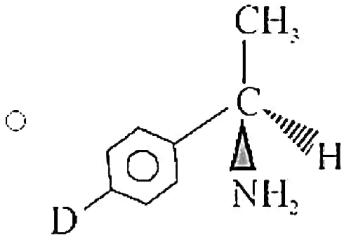
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Question No. 12 **(Only One Option Correct Type)**

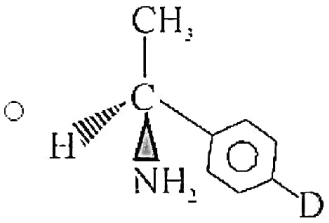


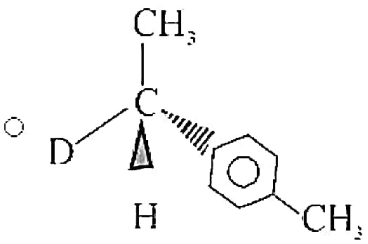
A)



B) 

C)



D) 

Question No. 13 **(Only One Option Correct Type)**

The e.m.f. of a cell corresponding to the reaction

$$Zn + 2H^+(aq) \longrightarrow Zn^{2+}(0.1M) + H_2(g)(1\ atm)$$

is 0.26 volt at $25^\circ C$. Calculate the pH of the solution at the hydrogen electrode.

$E^\circ_{Zn^{2+}/Zn} = -0.76$ volt and $E^\circ_{H^+/H_2} = 0$

- 8
- 9
- 1
- 2

Question No. 14 **(Only One Option Correct Type)**

The values of ΔH and ΔS for the reaction,
C (graphite) $+CO_2(g) \rightarrow 2CO(g)$ are 170 kJ and $170 JK^{-1}$, respectively. This reaction will be spontaneous at

- 910 K
- 1110 K
- 510 K
- 710 K

Question No. 15 **(Only One Option Correct Type)**

A mixture of methane and ethene in a molar ratio of x : y has an average molecular mass of 20 u. The mean molar mass when they are mixed in the molar ratio of y : x will be -

- 20 u
- 25 u
- 24 u
- 15 u

Question No. 16 **(Only One Option Correct Type)**

Match the following

	List I		List II
	(Group 13 element)		(metallic radius, pm)
A	Al	I	135
B	Ga	II	143
C	*In*	III	170
D	Tl	IV	167

The correct answer is

- A-I, B-II, C-IV, D-III
- A-II, B-I, C-IV, D-III
- A-III, B-IV, C-II, D-I
- A-II, B-I, C-III, D-IV

Question No. 17 **(Only One Option Correct Type)**

Which of the following statements are not correct?

- All $C-O$ bonds in CO_3^{2-} are equal but not in H_2CO_3
- All $C-O$ bonds in HCO_2^- are equal but not in HCO_2H
- $C-O$ bond length in HCO_2^- is longer than $C-O$ bond length in CO_3^{2-}
- None of these

Question No. 18 **(Only One Option Correct Type)**

The following homogeneous gaseous reactions were experimentally found to be second order overall.
1.) $2NO \rightarrow N_2 + O_2$
2.) $3O_2 \rightarrow 2O_3$
3.) $N_2O_3 \rightarrow NO + NO_2$
4.) $H_2 + I_2 \rightarrow 2HI$
Which of these are most likely to be elementary reaction that occur in one step?

- ○ 1 and 3
- ○ 1 and 4
- ○ 3 and 4
- ○ 1, 2 and 3

Question No. 19 **(Only One Option Correct Type)**

Reduction of hexose A (molecular formula $C_6H_{12}O_6$) with sodium borohydride gives compound B and C. Compound B is optically inactive, whereas compound C is optically active. Which of the following is compound A?

- ○ D-fructose
- ○ D-glucose
- ○ D-mannose
- ○ D-galactose

Question No. 20 **(Only One Option Correct Type)**

Which of the following is incorrect?

- ○ Molecular weight of $NaCl$ found by osmotic pressure measurements if half of the theoretical value.
- ○ Molecular weight of CH_3COOH in benzene found by cryoscopic methods is double of the theoretical value.
- ○ Osmotic pressure of 0.1 M glucose solution is half of that 0.1 M $NaCl$ solution.
- ○ Molecular weight of HCl found by any colligative property will be same in the aqueous solution and benzene solution.

Chemistry Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

How many statements given below are correct?

i) Pyrolusite on fusion with KNO_3 in NaOH gives K_2MnO_4.

ii) Potassium manganate (K_2MnO_4) on electrolytic oxidation gives $KMnO_4$.

iii) Potassium manganate (K_2MnO_4) is formed, when formaldehyde reacts with potassium permanganate in acidic medium.

iv) Potassium manganate (K_2MnO_4) is formed, when potassium permanganate is moderately heated with potassium hydroxide.

Question No. 2 **(Numerical Type)**

Find the quantum number 'n' corresponding to the excited state of He^+ ion, if on transition to the ground state that ion emits two photons in succession with wavelengths 108. 5 and 30. 4 nm.

Question No. 3 **(Numerical Type)**

The standard molar enthalpies of formation of cycohexane (l) and benzene (l) at $25^\circ C$ are -156 and +49 KJ mol^{-1} respectively. The standard enthalpy of hydrogenation of cyclo hexene (l) at $25^\circ C$ is -116 KJ mol^{-1} . Use these data to estimate the magnitude of the resonance energy of benzene. Mark answer in KJ per mol

Question No. 4 **(Numerical Type)**

Relative decrease in vapour pressure is 0.4 for a solution containing 1 mole AB in 3 mol H_2O, therefore AB is ionised at the extent of (mark answer in percentage)

Question No. 5 **(Numerical Type)**

pH of a buffer solution decreases by 0.02 units when 0.12 g of acetic acid is added to 250 mL of a buffer solution of acetic acid and potassium acetate at $27^\circ C$. The buffer capacity of the solution is β, find 10β.

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

Let $\vec{x}=\vec{a}+\vec{b}, \vec{y}=2\vec{a}-\vec{b}$, then the point of intersection of straight lines $\vec{r}\times\vec{x}=\vec{y}\times\vec{x}, \vec{r}\times\vec{y}=\vec{x}\times\vec{y}$ is

- $2\vec{b}$
- $3\vec{b}$
- $3\vec{a}$
- $2\vec{a}$

Question No. 2 **(Only One Option Correct Type)**

The ends of the base of an isosceles triangle at $(2,0)$ and $(0,1)$ and the equation of one side is $x=2$ then the orthocenter of the triangle is :

- $\left(\frac{2}{3},\frac{3}{5}\right)$
- $\left(\frac{5}{4},1\right)$
- $\left(\frac{3}{4},1\right)$
- $\left(\frac{4}{3},\frac{7}{2}\right)$

Question No. 3 **(Only One Option Correct Type)**

In an increasing geometric progression, the sum of the first and the last term is 99, the product of the second and the second last term is 288 and the sum of all the terms is 189. Then, the number of terms in the progression is equal to

- 5
- 6
- 6
- 7
- 8

Question No. 4 **(Only One Option Correct Type)**

The total number of local maxima and local minima of the function $f(x) = \begin{cases} (2+x)^3; & -3 < x \leq -1 \\ x^{\frac{2}{3}}; & -1 < x < 2 \end{cases}$ is

- 0
- 1
- 2
- 3

Question No. 5 **(Only One Option Correct Type)**

Coefficient of x^{n-1} in the expansion of, $(x+3)^n + (x+3)^{n-1}(x+2) \div (x+3)^{n-2}(x+2)^2 + \ldots\ldots \div (x+2)^n$ is.....

- ${}^{n \div 1}C_2(3)$
- $C_2^{n-1}(5)$
- ${}^{n \div 1}C_2(5)$
- $C_2^n(5)$

Question No. 6 **(Only One Option Correct Type)**

The area (in sq units) bounded by the curves $x = -2y^2$ and $x = 1 - 3y^2$ is____

- $\frac{2}{3}$
- 1
- $\frac{4}{3}$
- $\frac{2}{3}$

Question No. 7 **(Only One Option Correct Type)**

Suppose that a bag A contains n red and 2 black balls and another bag B contains 2 red and n black balls. One of the two bags is selected at random and two balls are drawn from it at a time. When it is known that the two balls drawn are red, if the probability that those two balls drawn are from bag A is $\frac{6}{7}$, then $n =$

- 6
- 4
- 8
- 7

Question No. 8 **(Only One Option Correct Type)**

An extreme value of $f(x) = \frac{4}{\sin x} + \frac{1}{1-\sin x}$ in $(0, \pi/2)$ is---

- 9
- 8
- $2/3$
- $-7/2$

Question No. 9 **(Only One Option Correct Type)**

If PQR is an equilateral triangle inscribed in the auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ and $P'Q'R'$ is the corresponding triangle inscribed within the ellipse, then centroid of the triangle $P'Q'R'$ lies at

- centre of ellipse
- focus of ellipse
- between focus and centre on major axis
- none of these

Question No. 10 **(Only One Option Correct Type)**

The number of complex numbers z such that $\arg(z + 3i) = \frac{\pi}{4}$ and $\arg(z - 2) = \frac{\pi}{3}$ is

○ 0

○ 1

○ 2

○ 3

Question No. 11 **(Only One Option Correct Type)**

Every curve represented by the general solution of $\frac{dy}{dx} = \frac{x \log x}{y^3 e^{y^2-5}}$ cuts every curve represented by the general solution of $\frac{dy}{dx} + \frac{y^3 e^{y^2-5}}{x \log x} = 0$ at angle θ. Then, $4\theta - \frac{\pi}{2}$ =----

○ $\frac{\pi}{2}$

○ 2π

○ $\frac{3\pi}{2}$

○ π

Question No. 12 **(Only One Option Correct Type)**

If $f(x) = \max\left\{\frac{x}{n}, |\sin\pi x|\right\}$, $n \in N$ has maximum points of non-differentiability for $x \in (0, 4)$, then n cannot be

○ 4

○ 2

○ 5

○ 6

Question No. 13 **(Only One Option Correct Type)**

The length of the common chord of the circles of radii 15 and 20, whose centres are 25 unit of distance apart, is

○ 12

○ 18

○ 24

○ 16

Question No. 14 **(Only One Option Correct Type)**

Let α, β are the roots of the quadratic equation $2x^2 - 5x + 1 = 0$. If $S_n = (\alpha)^{2n} + (\beta)^{2n}$ then find the value of $\frac{4S_{2021}+S_{2019}}{S_{2020}}$.....

- 27
- 21
- 24
- 32

Question No. 15 **(Only One Option Correct Type)**

Let $f : \mathrm{R} \to \mathrm{R}$ be a function such that $f\left(x\right) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$, where $[x]$ denotes the greatest integer function. If $I = \int_{-1}^{2} \frac{x.f(x^2)}{f(x+1)+2}\, dx$, then the value of $2I =$_____:

- 1
- 1.5
- 0.5
- 2.5

Question No. 16 **(Only One Option Correct Type)**

If n satisfies the relation ${}^{n+5}P_{n+1} = \frac{11(n-1)^{n+3}}{2} P_n$, then maximum number of points of intersection of ' n ' distinct straight lines is

- 18
- 28
- 21
- 10

Question No. 17 **(Only One Option Correct Type)**

Let $a_1 = 0$ and $a_1, a_2, a_3, \ldots\ldots, a_n$ be real numbers such that $|a_i| = |a_{i-1} + 1|$ for all i, then the A.M. of the numbers $a_1, a_2, a_3, \ldots\ldots, a_n$ has the value A where :

- $A < -\frac{1}{2}$
- $A < -1$

- $A \geq -\frac{1}{2}$
- $A = -2$

Question No. 18 **(Only One Option Correct Type)**

If $x\sqrt{1+y}+y\sqrt{1+x}=0$, then $\frac{dy}{dx}$ is equal to----

- $\frac{1}{(1+x)^2}$
- $-\frac{1}{(1+x)^2}$
- $\frac{1}{1+x^2}$
- $\frac{1}{1-x^2}$

Question No. 19 **(Only One Option Correct Type)**

If $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\frac{2x}{1+x^2}\,dx = k\int_0^{1/\sqrt{3}} \frac{x^4}{1-x^4}\,dx$ then 'k' equals

- π
- 2π
- 2
- 1

Question No. 20 **(Only One Option Correct Type)**

The number of matrices of A for which the system of linear equations $A\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$ has a unique solution, is---

- less than 4
- atleast 4 but less than 7

- atleast 7 but less than 10
- atleast 10

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

The mean and standard deviation of 100 observations were calculated as 40 and 5.1, respectively by a student who took by mistake 50 instead of 40 for one observation, then the correct standard deviation is

Question No. 2 **(Numerical Type)**

The direction cosines (l, m, n) of two lines satisfy $3l + m + 5n = 0$, $6mn - 2nl \div 5lm = 0$. If ' θ ' is the angle between those two lines then the value of $\left[\frac{6\tan^3\theta - 5\sec\theta}{10\pi}\right]$ where [.] denotes greatest integer function is

Question No. 3 **(Numerical Type)**

Let $S = \{1, 2, 3, 5, 8, 13, 21, 34\}$ and $A = \{x_1, x_2\}$, a two elements subset of S. Find $\Sigma\max(A)$. (where the sum is taken over all such subsets)

Question No. 4 **(Numerical Type)**

If p is a positive integer and f be a function defined for positive numbers and attains only positive values such that $f(xf(y)) = x^p y^4$ then find p.

Question No. 5 **(Numerical Type)**

Let $f(x) = \begin{bmatrix} x^2 + a & , 0 \le x < 1 \\ 2x + b & , 1 \le x \le 2 \end{bmatrix}$ and $g(x) = \begin{bmatrix} 3x + b, & 0 \le x < 1 \\ x^3 & , 1 \le x \le 2 \end{bmatrix}$

If derivative of $f(x)$ w.r.t. $g(x)$ at $x = 1$ exists, and has value equal to m then the value of $|a \div b + 6m|$ is equal to

(ANSWER KEY)

Physics

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

Chemistry

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

Mathematics

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]